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Traveling or Jiggling: Particle Motion Modes and Their Relative Contribution to Bed-Load Variables

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Key Points:

- We distinguish "transport" (T) and "non-transport" (NT) states of particle motion and analyze them separately
- The NT state involves isotropic jiggling of particles that do not contribute to the sediment discharge
- The NT movements impact considerably the expected values of number of moving particles and velocity

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Abstract The motion state of a particle is a crucial aspect of sediment transport problems. In this paper, we conceptualized three states: stillness, "transport", and "non-transport", considering that not all the particle motions contribute significantly to the mean sediment transport rate. Starting from a data set of bed-load particle tracks obtained from particle tracking velocimetry, we removed the bias from experimental uncertainty and applied one-dimensional, instantaneous, and non-parametric criteria for distinguishing the states. We described the kinematics of particles in transport and non-transport states, presenting some sample trajectories and the distributions of particle velocity and acceleration. While the transport state presents a clear distinction between stream-wise and transverse particle velocity, the non-transport state is related to isotropic particle jiggling, and does not significantly contribute to the bed-load rate. Vice-versa, the particle motions in the non-transport state are relevant for other summary indicators of the transport process, such as the mean number of moving particles and mean particle velocity. We discuss how applying the proposed non-parametric criterion for state separation is beneficial compared to parameter-dependent alternatives available in the literature. Finally, we provide an outlook on possible applications of our concept for the investigation of other sediment transport processes (incipient motion, solid-fluid interface, creeping flow).

Plain Language Summary Bed-load is the transport of sediment particles close to a channel bed. When the sediment transport rate is low, individual particles can be alternatively in motion or stillness. Quantitative determination of relevant quantities (number of moving particles, mean velocity, and others) may require a definition of criteria based on which a particle should be considered in motion or not at a given time instant (an issue far from being straightforward). Furthermore, particles can jiggle around fixed positions without contributing to downstream transport. In this work, we proposed criteria for labeling the stillness and motion states. The latter is split further into two states: "transport" (associated with neat downstream motion) and "non-transport" (associated with jiggling). Eventually, we characterized the properties of the states of motion and their contributions to global indicators of the bed-load process. We discuss the appropriateness of our definitions compared to alternatives available in the literature and the usefulness of state separation for bed-load processes investigation.

1. Introduction

The transition between two states—motion and rest—is one of the keystones of bed-load processes, where it is common experimental evidence that particle motion is intermittent rather than continuous (Ancey et al., 2008). Furthermore, as discussed by, for example, Furbish et al. (2012) and Ballio et al. (2018), the sediment transport rate can be expressed either in a "flux form" or in an "entrainment form"; in the first case, it is given by the product of a particle velocity and a particle concentration, where both quantities are space-averaged (Kalinske, 1947) and, in the second case, by the product of a double-averaged entrainment rate and a mean hop length (Einstein, 1937). Nikora et al. (2001, 2002) proposed three spatial and temporal ranges for bed-load motion: the local (between two successive collisions of a particle with the bed), intermediate (particle trajectories between two successive periods of rest), and global ranges (many intermediate trajectories). Many scholars focused on studying bed-load kinematics at the intermediate trajectory scale (Lajeunesse et al., 2010; Liu et al., 2019; Roseberry et al., 2012), investigating hop length and duration or particle resting time. Finally, quantities related to single particle motion also may be used to derive others (entrainment and deposition rates, particle activity) that are commonly involved in mathematical models of the bed-load transport process (Ancey et al., 2008; Charru et al., 2004; Fan et al., 2016). All the mentioned approaches require a clear definition of what stillness and motion are meant to be, as well as of the transition between the two states.

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Bed-load can be thought to comprise three states instead of two, splitting the state of motion into two different states. The first one is given by particles that, at a certain time, are displaced downstream (by saltation, rolling, or sliding); the second one corresponds to particles that jiggle around a certain position without a neat displacement after some time. Many names have been adopted for the jiggling state: "jiggle" (Wu et al., 2020), "rocking motion before the actual hop" (Liu et al., 2019), "shaking" (Shim & Duan, 2019), particles that "wiggled back and forth within their pockets" (Fathel et al., 2016), "rocking back and forth, or jiggling but not actively moving" (González et al., 2017), "wobbling but not moving" (Heays et al., 2014). Here, we chose the term "jiggling" for this motion; furthermore, we made a subsequent step and, like, for example (Salevan et al., 2017), we named the two motion states in relation to their contribution to the bed-load flux, distinguishing them as "transport" (T) and "non-transport" (NT).

Scholars proposed several operational definitions of the transition between motion and rest in the literature. Some of them are based on classifying a particle's state based on a cut-off value for one-dimensional (Ancey & Heyman, 2014; Lajeunesse et al., 2010; Roseberry et al., 2012), two-dimensional (Böhm et al., 2004; Sechet & Guennec, 1999) or even three-dimensional (González et al., 2017) particle instantaneous velocity. Other definitions are based on properties of particle motion and rest over more than one instant: Cecchetto et al. (2018), for example, prescribed a resting time greater than 0.1 s and a hop length longer than 1 grain diameter for a particle to be considered at rest or in a motion, respectively. Martin et al. (2012) stated that "a resting particle was considered entrained when it moved a complete particle diameter, while a mobile particle was considered distrained when it failed to move more than 1 d (here d is the particle size) within a second." Liu et al. (2019) considered only "long particle trajectories, with integrated displacement over 10 d and experiencing at least one step-stop-step sequence of motion and rest." Eventually, some authors used second moments of particle position; for example, Seizilles et al. (2014) labeled a particle as moving "if the standard deviation of its position over four successive pictures is larger than 0.1 d."

The definitions of motion and the experimental investigations cited above have been aimed at excluding NT motions, considering the jiggling particles as if they were at rest. Indeed, Cecchetto et al. (2018) and Liu et al. (2019) stated that their criteria of motion were adopted to exclude the non-transport motion state; Fathel et al. (2016) did not track the jiggling particles. By contrast, just a few authors have focused on the NT state of particle motion. Salevan et al. (2017) concluded that it is impossible to separate the T and NT states based on a threshold velocity and, therefore, proposed a mixed model for the probability density function (PDF) of the velocity accounting for both motion states. Liu et al. (2019) also introduced three states: "moving" and "active waiting" (similar to T and NT, respectively), and "deep waiting" (stillness).

There is merit in studying the phenomenological properties of both motion states (T and NT), as they may differently contribute to multiple facets of the bed-load process. Moreover, the properties of the NT state have been just limitedly analyzed before. The research questions of the present manuscript are thus: (a) what are the kinematic properties of the two states of particle motion? (b) what is the relative contribution of the T and NT states on variables commonly used in sediment transport problems? As already highlighted above, answering these questions requires a preliminary definition of how one should distinguish T and NT, to make the NT motions irrelevant to the mean sediment transport rate. Therefore, we also present a new, non-parametric criterion for labeling a particle state at any sampling time during the process; the non-parametric criterion is advantageous compared to others used in the literature (e.g., those based on cut-off velocity values that are threshold-dependent). The analysis is anchored to particle tracks from an earlier experimental campaign with weak bed-load.

2. Experimental Set-Up and Methods

The experiments analyzed in this manuscript were carried out by Campagnol et al. (2012) at the Hydraulics Laboratory of the Politecnico di Milano, employing a pressurized duct (5.80 m long, 0.40 m wide, 0.11 m high). In the downstream part of the flume, they filled a recess section (2 m long and 0.4 m deep) with polybutylene terephthalate (PBT) quasi-spheres with an equivalent diameter $d = 3$ mm and a density $\rho_s = 1{,}270$ kg/m^3. They glued the same particles on a series of plates positioned in the upstream reach (3.80 m long). Campagnol et al. (2013, 2015) give a detailed description of the experimental set-up. In the present study, we used the results of three experiments performed at $u_*/u_{*c} = 1.11$, 1.22, and 1.30 (u_* is the shear velocity of the flow and u_{*c} is its threshold value for bed-load).

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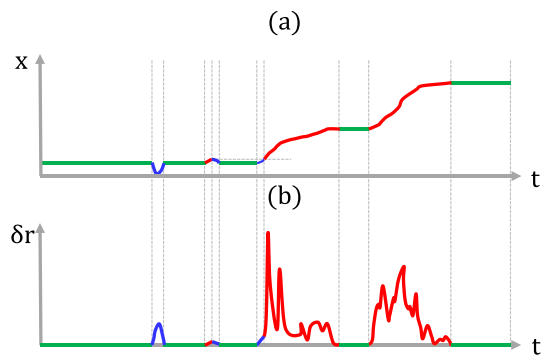

Figure 1. Example of a particle trajectory: temporal evolution of (a) stream-wise position and (b) absolute displacement δr between two frames. Lines are colored according to particle states: at rest (green); state T (red); state NT (blue).

Bed-load particle motion was investigated using particle tracking velocimetry (PTV) after taking videos with a digital camera (frame resolution and rate were 970 × 700 pixels and 32 fps, respectively). The framed area had a stream-wise length of 49.8 cm and a transverse one of 36.5 cm; the duration of the videos was approximately 50 s.

The sediment was a mixture of black and white particles (95% and 5%, respectively): we tracked only the white ones. We measured particle trajectories using the Streams Software (Nokes, 2012) following procedures already implemented by Radice et al. (2017). After background removal, we converted grayscale frames into binary images using a single threshold monochrome algorithm. We fixed the threshold intensity at 45 (out of 255). We set thresholds for minimum and maximum diameter of particle spots in binary images at 1.5 and 4.5 mm. The range was quite large compared to d because variable spot sizes could be obtained based on how light was reflected by any particle. Then, we obtained particle tracks using a two-dimensional search algorithm with global optimization; the searching window for a particle displacement within 1/32 s was ±10 and ±10 mm in the stream-wise and transverse directions, respectively.

3. Framework for Analysis

3.1. Conceptual Definition of Motion States

In the present study, we restrict our attention to one-dimensional (on average) flows. However, transverse motion is also possible: thus, we take x and y as the stream-wise and transverse coordinates, respectively. During a small time interval δt, a particle performs a displacement with components $\delta x(t) = x(t) - x(t - \delta t)$ and $\delta y(t) = y(t) - y(t - \delta t)$, and magnitude $\delta r(t) = \sqrt{(\delta x(t))^2 + (\delta y(t))^2}$. The quasi-instantaneous (i.e., mean within δt) velocity components in the stream-wise and transverse directions are $u(t) = \delta x(t)/\delta t$ and $v(t) = \delta y(t)/\delta t$; the velocity magnitude is $V(t) = \sqrt{(u^2(t) + v^2(t))}$. Finally, the components of the quasi-instantaneous particle acceleration are $a_x(t) = [u(t) - u(t - \delta t)]/\delta t$ and $a_y(t) = [v(t) - v(t - \delta t)]/\delta t$.

At any time, a particle may be in three states: stillness, T state of motion, NT state of motion. Figure 1 shows an idealized global trajectory. A particle is said to be at rest, at time t, if its absolute displacement is null (in green), whereas it is moving if the absolute displacement is different from zero (red and blue). Furthermore, a moving particle is said to be in T state if, at time t, it occupies a stream-wise position that has never occupied previously while moving downstream (we will provide a formal definition later). If not, we label its motion state as NT. In Figure 1 the states T and NT are depicted in red and blue, respectively. The total stream-wise distance traveled in the T state of motion equals the total stream-wise distance traveled by the particle.

3.2. Operational Issues

Figure 2 summarizes the algorithm used for labeling the motion state of a particle. Besides implementing the conceptual definition described above, the algorithm starts accounting for experimental uncertainties. Even if, in principle, one could apply the definitions to the tracks obtained with Streams, not all the displacements measured by the software actually corresponded to real ones. Any variation of the shape of a particle spot in a frame caused a corresponding variation of the position in the particle centroid, resulting in a displacement detected by the PTV algorithm. There were two main causes for this error: (a) small changes of light intensity produced a deformation of the particle; (b) the spot of another particle could partially hide (in case of a black one) that of a white still particle or overlap with it (in case of a white one) when they became sufficiently close to each other, resulting in a misattribution of NT state of motion to a still particle.

To address these problems, we manually selected 30 particles (for each experiment) remaining still along a movie and analyzed the corresponding time series of images. Furthermore, for the time series of the still particles (Figure 2a), we excluded all the instants where we detected overlap with a white particle or hiding by a black particle (shaded area in Figure 2b). In this way, we built a sample of data for particles that a human eye would

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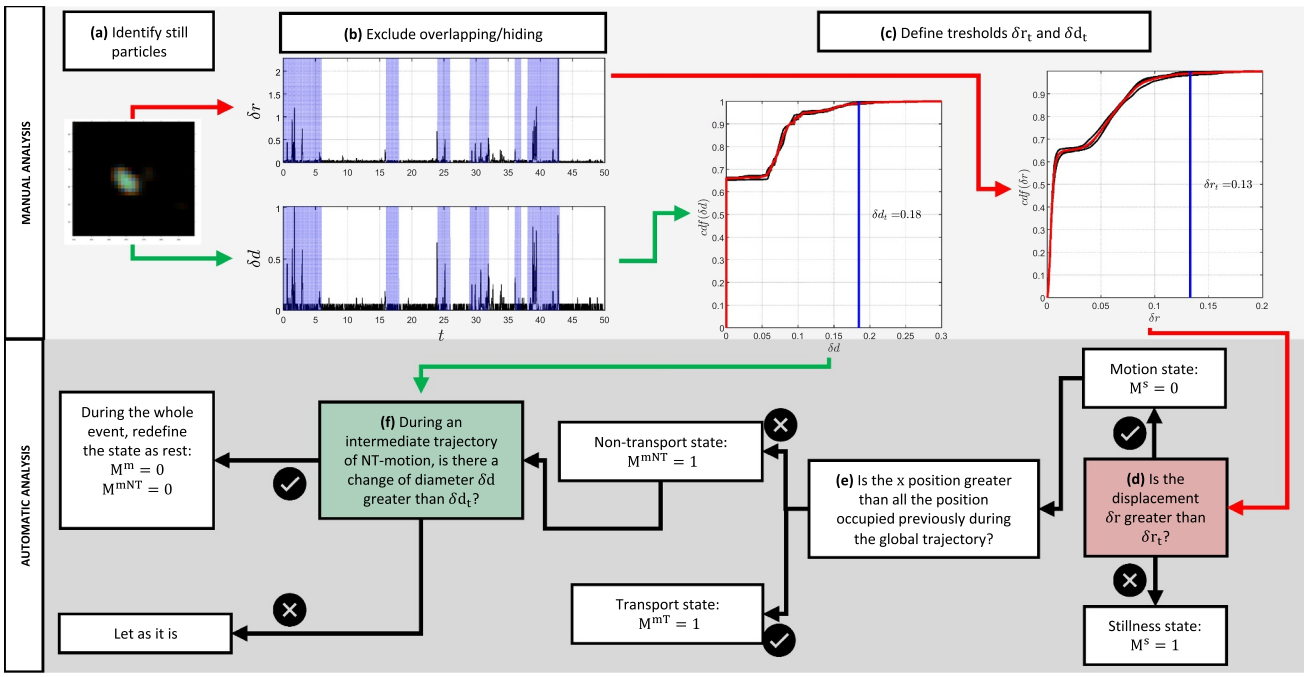

Figure 2. Schematic representation of the algorithm used for states' labeling.

soundly consider still while watching a movie. To overcome the first issue above (particle deformation due to changes of intensity, with a consequent displacement of its centroid), we computed the frequency distribution function of displacements for these still particles: for the three experiments the distributions (Figure 2c) were similar and, merging the three data sets, 99% of the displacements were smaller than 0.13 mm. Therefore, we took $\delta r_t = 0.13$ mm as the minimum measurable displacement; we considered displacements lower than this threshold to be actually null, with the particle in a state of stillness. Since we removed any displacement lower than δr_t from the data set, velocities lower than $\delta r_t \times \delta t = 4$ mm/s (in magnitude) were absent in the final data set. However, velocity components lower than this value could be present. In principle, one could think to do this job in an opposite manner, finding a minimum δr for manually identified moving particles. In practice, this was impossible for human eyes, while we identified still particles more reliably. Concerning the second issue (particle hiding or overlap), we considered the same sample mentioned above and computed the frequency distribution of the diameter change δd (Figure 2c). Distributions for the three experiments were again similar and, for the experiments together, 99% of the change values were smaller than $\delta d_t = 0.18$ mm. Any diameter change bigger than δd_t was thus reasonably attributed to a crossing. The measured centroid's position before and after hiding/overlapping occurrence was the same, mimicking an NT state trajectory. For this reason, we took motions previously labeled as in NT state back to stillness if they presented spot size changes larger than δd_t.

Our experiments were performed by Campagnol et al. (2013) with the combination of focus area, image resolution, and frame rate that was allowed at that time by the experimental setup. The thresholds just introduced are likely to depend on operational conditions. Given the impossibility of modifying the image resolution and increasing the frame rate, we present in Appendix A a sensitivity analysis of the threshold for lowering the frame rate (that, for the sake of the exercise, was obtained by subsampling the already measured particle tracks). Generally, the thresholds increase for lower frequency. As a complement to this analysis, Appendix A also shows how the frequency distribution of measured particle velocity (described later in the manuscript) changes with the measuring frequency.

3.3. Clipping Functions

Based on the conceptual and operational considerations above, we grounded the recognition/labeling of a particle state on the following clipping functions (as described in Figures 2d and 2e): we used $M^s(i, t)$ to recognize stillness based on a minimum displacement (i is a particle counter);

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$$M^s(i,t) = \begin{cases} 1, & \delta r(t) \le \delta r_t \\ 0, & \delta r(t) > \delta r_t \end{cases} \tag{1}$$

while we used $M^{mT}(i, t)$ for the T state of motion based on a position criterion, mirroring that proposed by Campagnol et al. (2015):

$$M^{mT}(i,t) = \begin{cases} 1, & x(t) > \max\limits_{0 \le \tau \le t}[x(\tau)] \\ 0, & x(t) \le \max\limits_{0 \le \tau \le t}[x(\tau)] \end{cases} \tag{2}$$

Finally, we used a function $M^{mNT}(i, t)$ for the NT state of motion. Given that, at any time, a particle can occupy only one of the three states, it follows that $M^{mNT}(i,t) = 1 - M^{mT}(i,t) - M^s(i,t)$. As said, after applying Equations 1 and 2, we turned some NT motion events into stillness events if excessive variation ($\delta d(t) > \delta d_t$) of particle diameter was observed (Figure 2f).

We also introduced a clipping function used to obtain the sediment transport rate to label particle crossings of a transverse line. Considering a time interval δt, we computed the sediment transport rate through a transverse line at time t starting from the volume of sediment crossing the line during the time period from $t - \delta t$ to t (Ballio et al., 2014). Thus, for a reference transverse line with $x = x^*$, the clipping function $M_Q(i, t)$ stated whether particle i crossed the line and if it crossed it moving forward or backward during the time interval $[t - \delta t, t]$:

$$M^Q(i,t) = \begin{cases} +1, & \text{if } x(t-\delta t) \le x_0,\ x(t) > x_0, \\ -1, & \text{if } x(t-\delta t) \ge x_0,\ x(t) < x_0, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

3.4. Definition of Motion Variables

We provided expressions here for some Eulerian variables commonly used in bed-load modeling. Let $N(t)$ be the total number of particles in the framed area at time t, $N^m(t)$ the number of moving particles, $N^{mT}(t)$ the number of particles moving with state T, and $N^{mNT}(t)$ the number of those moving with state NT. In this work, we determined the variables as follows:

$$\begin{cases} N^m(t) &= \sum_{i=1}^{N(t)}(1 - M^s(i,t)) = \sum_{i=1}^{N(t)}\left[M^{mT}(i,t) + M^{mNT}(i,t)\right] \\ N^{mT}(t) &= \sum_{i=1}^{N(t)} M^{mT}(i,t) \\ N^{mNT}(t) &= \sum_{i=1}^{N(t)} M^{mNT}(i,t) \end{cases} \tag{4}$$

Furthermore, the total number of particles that crossed the transverse line at $x = x^*$ at time t was given by:

$$N_Q(t) = \sum_{i=1}^{N} M^Q(i,t)(1 - M^s(i,t)) = \sum_{i=1}^{N} M^Q(i,t)\left[M^{mT}(i,t) + M^{mNT}(i,t)\right] \tag{5}$$

We also subdivided $N_Q(t)$ based on the two states T and NT using the following equations:

$$\begin{cases} N_Q^T(t) &= \sum_{i=1}^{N(t)} M^Q(i,t) M^{mT}(i,t) \\ N_Q^{NT}(t) &= \sum_{i=1}^{N(t)} M^Q(i,t) M^{mNT}(i,t) \end{cases} \tag{6}$$

Eventually, we introduced the sediment kinetic energy per unit mass, k, which could also be decomposed by separating the T and NT states:

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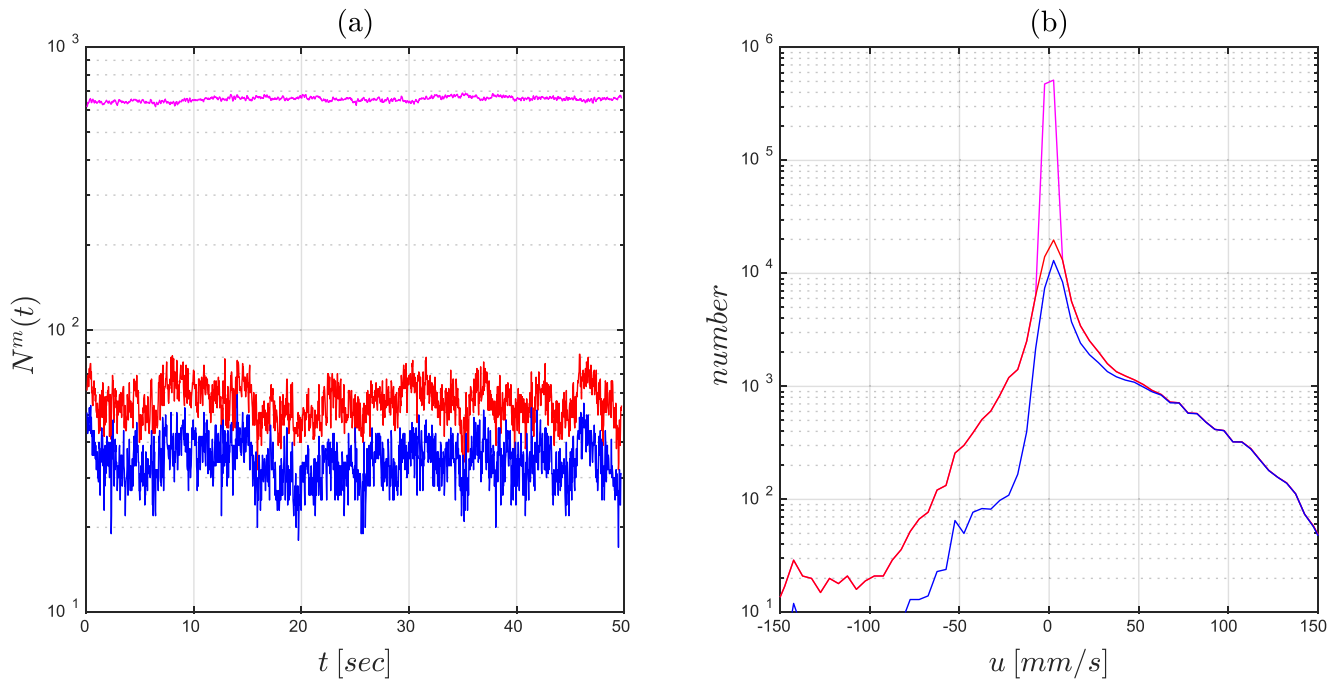

Figure 3. (a) Time series of N^m; (b) Histogram of the quasi-instantaneous stream-wise velocity u. Without any filter: $\delta r_t = 0$; $\delta d_t = \infty$ (magenta). With the small displacement filter only: $\delta r_t = 0.13$ mm; $\delta d_t = \infty$ (red). With both filters: $\delta r_t = 0.13$ mm; $\delta d_t = 0.18$ mm (blue). Both panels refer to $u_*/u_{*c} = 1.22$.

$$\begin{cases} k(t) &= \frac{1}{2}\sum_{i=1}^{N(t)} V(i,t)^2 (1 - M^s(i,t)) = \frac{1}{2}\sum_{i=1}^{N(t)} V(i,t)^2 \left[M^{mT}(i,t) + M^{mNT}(i,t)\right] \\ k^{sT}(t) &= \frac{1}{2}\sum_{i=1}^{N(t)} V(i,t)^2 M^{mT}(i,t) \\ k^{sNT}(t) &= \frac{1}{2}\sum_{i=1}^{N(t)} V(i,t)^2 M^{mNT}(i,t) \end{cases} \tag{7}$$

After defining relevant variables, we quantified how their determination depends on the operational criteria introduced above. For example, Figure 3 shows the effect of the filters δr_t and δd_t on the number of moving particles (we show the time series) and the particle stream-wise velocity (we represent the histogram). In magenta, we depict the results considering a particle in motion if its absolute displacement is non-null. In red, we show the results using Equation 1 to separate motion and stillness: the number of moving particles decreases by one order of magnitude and the central peak of the velocity histogram is strongly reduced. Then, we also apply the filter on the change of diameter δd_t, obtaining the blue curve: the number of moving particles is further reduced, and many negative velocities are removed.

4. Results

4.1. Separation Among the States of Motion

Figure 4 depicts intermediate trajectories obtained for particles in states T and NT. Consistently with the aim that motivated the separation between these states, trajectories of particles in state T correspond to neat motions in the stream-wise direction (Figure 4a), with hop lengths (the distances covered from start to halt) as large as several particle diameters; furthermore, for this state, the displacements in the stream-wise direction are much larger than those in the transverse one, that are however not negligible due to the interactions of the moving particles with the bed roughness and the flow turbulence. In contrast, the intermediate trajectories of particles in the NT state (Figure 4b) are significantly longer than our threshold for the instantaneous displacement δr_t but their projections along x and y are shorter than 1 particle diameter due to the particles

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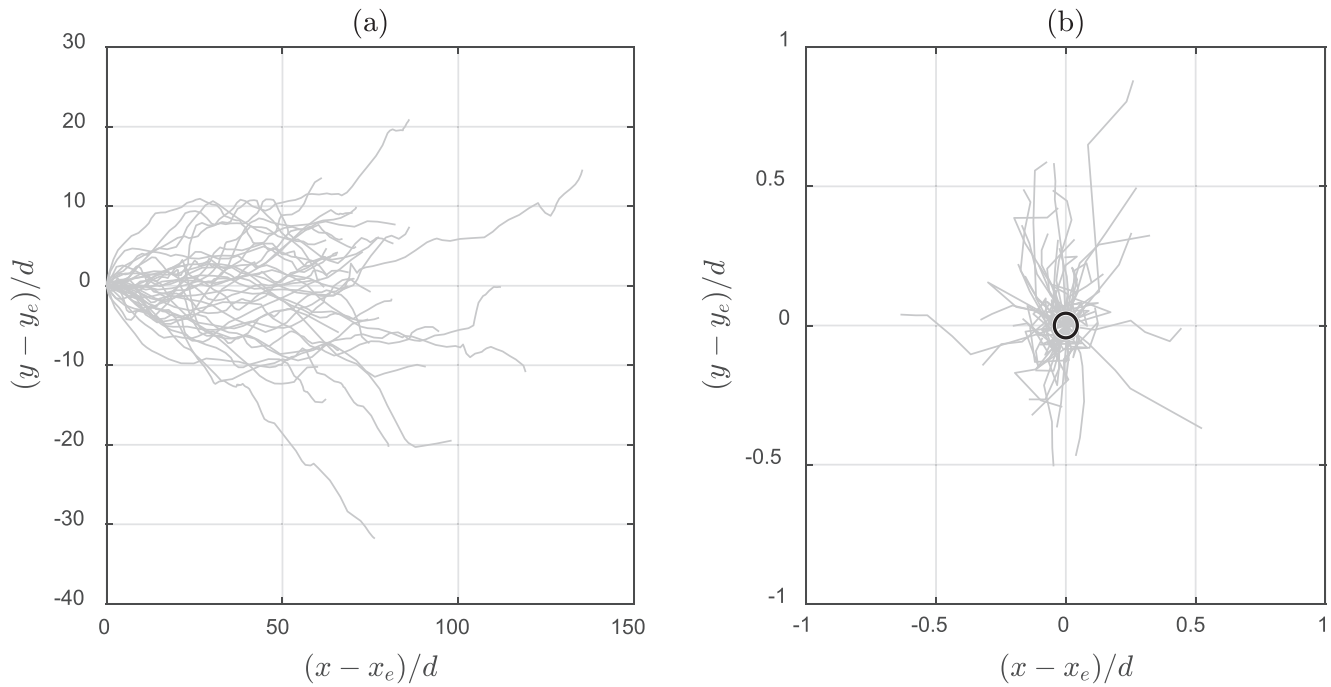

Figure 4. Example of intermediate trajectories from the run $u_*/u_{*c} = 1.30$ for (a) T state (b) NT state, with x_e and y_e as the coordinates of the location at which the movement started. In (b) we also show a circle with radius $\delta r_t = 0.13$ mm. On average, the time spent in motion is 0.1607 and 0.0408 s for the T and NT intermediate trajectories, respectively.

moving along very curled paths. This behavior is in line with the idea that a particle, during a hop in NT state, jiggles within a pocket (whose size is similar to the grain diameter) without leaving it, and it is similar to the kinematics observed by, for example, Celik et al. (2010) employing a photodetector to quantify particle entrainment.

Figure 5a shows the cumulative number of particles crossing a line placed at mid-length of the investigation area, which represents a cumulative volume of the sediment flux: the sediment transport is, indeed, almost entirely due to the T state, while the contribution of the NT state is negligible. Figure 5b shows instead the instantaneous value of N_Q: the contribution of state T (red) is always positive, while the one of NT (blue) can be either positive or negative. In our time series, $N_Q^{NT}(t)$ is mostly zero, but it never happens that, at the same time t, a forward and a backward motion are measured. However, sometimes a backward motion is followed by a forward motion almost immediately. Considering NT, one could expect a cumulative null value but, in our case, the net contribution to solid flowrate is negative; we explain this finding considering that a backward velocity is naturally appropriate for a motion in NT state, while it is less straightforward to separate forward motions.

Figures 6a–6d shows frequency distributions of velocity and acceleration for particles in the T and NT states. Considering the T state, the distribution of the velocity in the flow direction obviously exists only for positive values, while that of the transverse velocity is symmetric and centered at zero; Figure 6a thus proves an anisotropic behavior of state T. Instead, the distributions of the acceleration of the T state, shown in Figure 6b, are symmetric and centered at zero, which is a consequence of the steadiness of the flow. Nevertheless, the extreme values are more frequent for the acceleration in the flow direction than for the transverse acceleration. Particles in the NT state of motion do not have a preferential direction of motion while they jiggle, as the PDFs of the instantaneous stream-wise and transverse velocity components are similar to each other (Figure 6c); the distribution of u for the NT state is skewed toward negative values, reflecting the negative contribution to the sediment transport rate spotted in Figure 5a. The distribution of the quasi-instantaneous acceleration (Figure 6d) also shows no preferential direction, in addition to steadiness.

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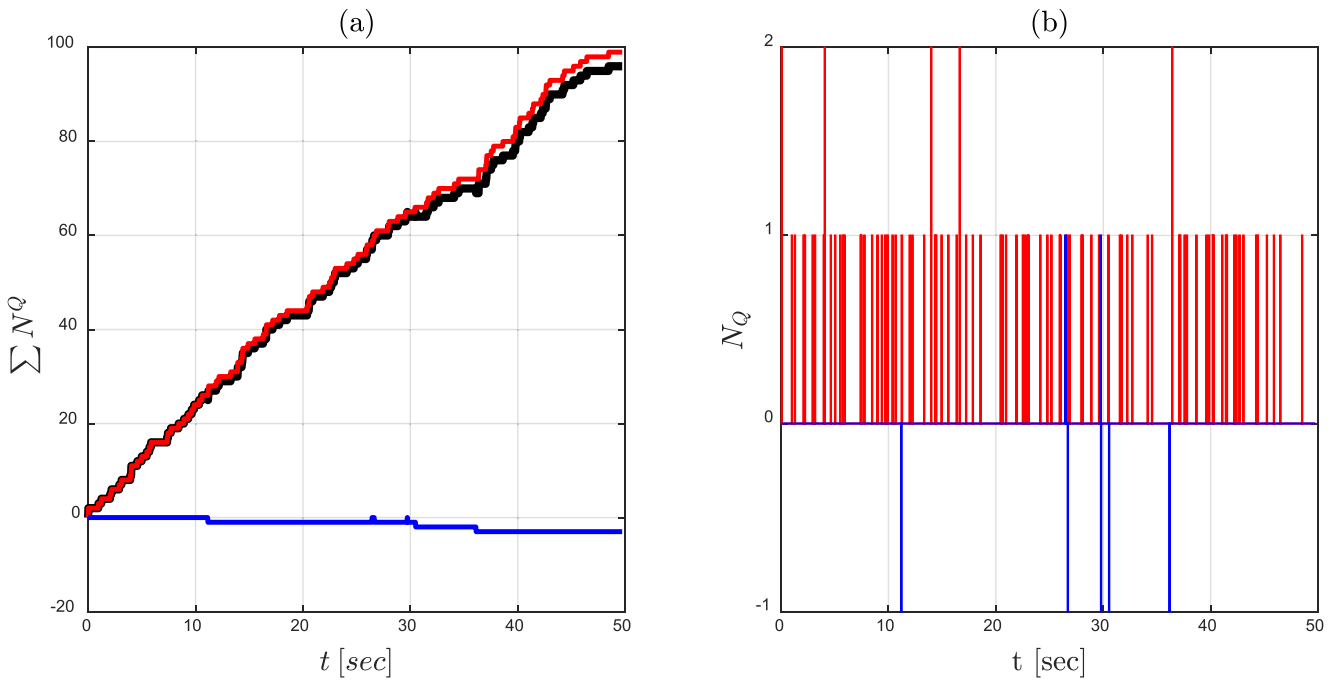

Figure 5. Cumulative number of particles crossing a line: all motions (black), NT state (blue) and T state (red). (b) Number of particles crossing a line: NT state (blue) and T state (red). The panels refer to $u_*/u_{*c} = 1.30$.

4.2. Instantaneous Stream-Wise Particle Velocity

Figure 7 shows the quasi-instantaneous stream-wise particle velocity histograms for all the transport conditions. The distributions without separating the motion states present a clear peak around zero and are positively skewed to the right. When the states of motion are separated, since our definitions attribute all the negative velocities to the particles in the NT state, the black and blue lines coincide for $u < 0$; furthermore, particles in this state jiggle around a fixed position, thus the corresponding motion velocities have a positive counterpart of the negative ones, resulting in an almost symmetric histogram. On the other hand, the T state is characterized by positive velocities, with a right tail of the histogram much longer than that of the particle velocity in the NT state; the mode of the distribution is at the lowest values.

A model for the probability distribution function of the velocities of all the moving particles was recently put forth by Salevan et al. (2017) then followed by Galanis et al. (2021). Their model and our experimental findings generally show good agreement (full details are furnished in Appendix B). One of the innovations of the present study is the ability to distinguish between motions in the T and NT states, providing the concept of Salevan et al. (2017) and Galanis et al. (2021) with empirical support for the first time.

4.3. Contribution of the Motion States to Bulk Variables

We present and discuss the effect of the two states of motion on six significant bulk variables using Table 1, where we report the values of the following variables: the cumulative number of particles that crossed a reference transverse line $\sum N_Q$; the total Σspace traveled in the stream-wise and transversal direction, $\sum \Delta_x$; and $\sum \Delta_x$; the mean kinetic energy per unit mass $\bar{k}$; the mean number of moving particles, $\overline{N_m}$; and the mean stream-wise velocity, $[u]$. We provide the values considering the motion states together and for the T and NT states separately, finally computing (for all quantities but the mean stream-wise velocity) the relative contributions of the two states to the totals.

For both motion states, all the variables increase for increasing shear velocity. The values obtained separating the T and NT states confirm that the latter has negligible contribution to $\sum N_Q$, generalizing the results of Figure 5a and supporting the method introduced in this study for state separation. Considering the stream-wise displacement, the NT motions amount to 16% as a maximum in our tested range of transport conditions. We also note that

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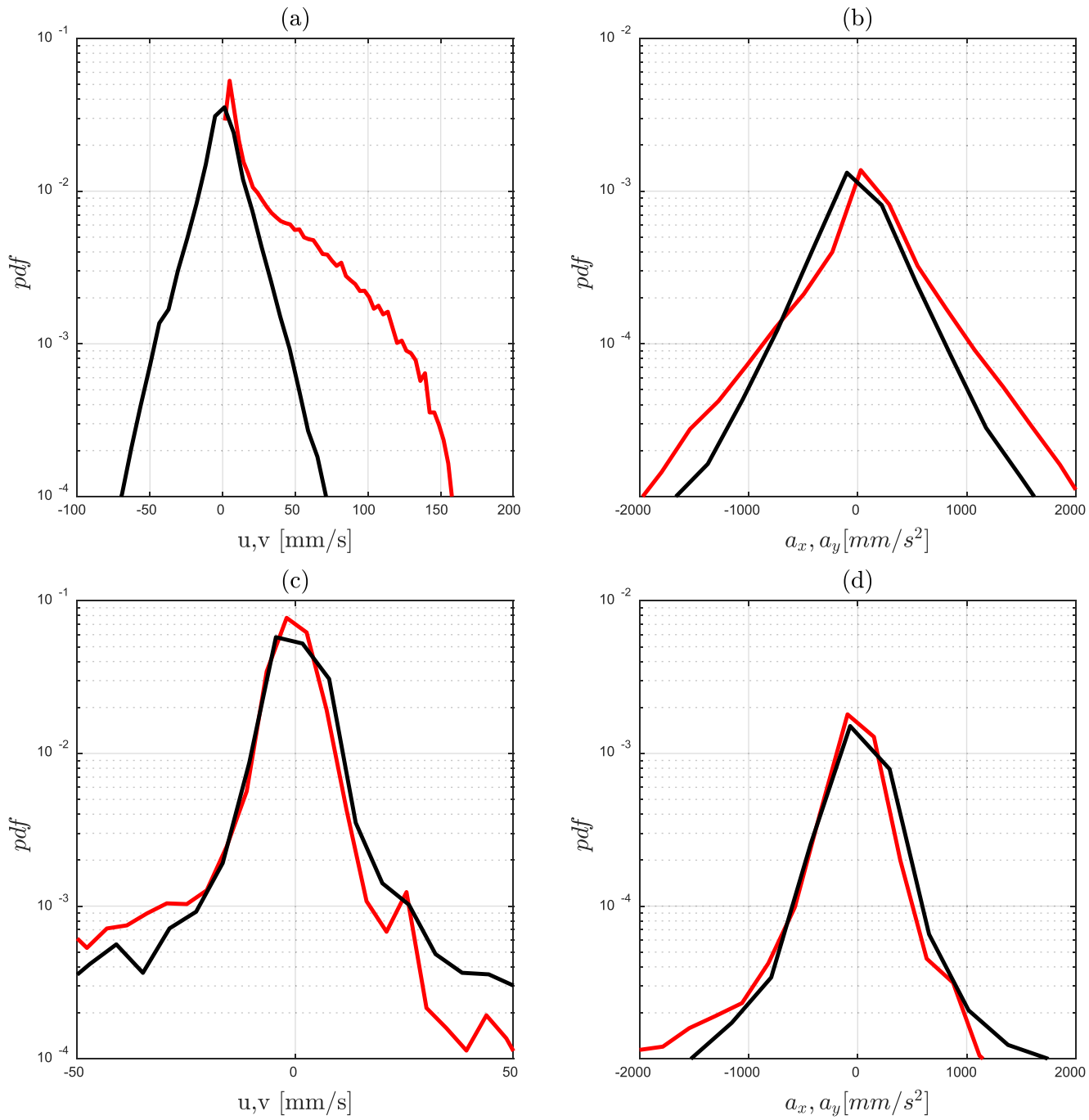

Figure 6. Observed probability distribution function of: u^T (red); v^T (black). (b) Observed probability distribution function of a_x^T (red); a_y^T (black). (c) Observed probability distribution function of: u^{NT} (red); v^{NT} (black). (d) Observed probability distribution function of a_x^{NT} (red); a_y^{NT} (black). The panels refer to $u_*/u_{*c} = 1.30$.

considering the absolute value of the displacement does not affect the motions in state T while it increases the indicator's value for those in state NT. Furthermore, the NT motions are quite significant for the transverse direction, where the absolute value increases the indicator value for both motion types. For both directions, the relative contribution of the NT motions is larger for lower transport intensities. The contribution of the NT state to the kinetic energy is similar to that to the stream-wise displacement and is not negligible as, for our experiments, it is around 10%, with a higher percentage for weaker flow conditions.

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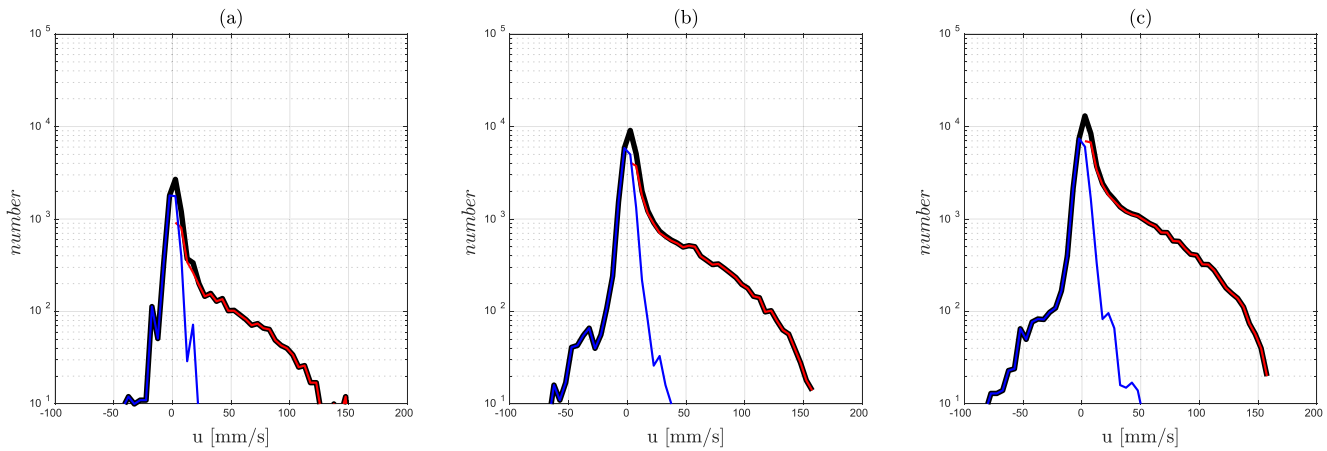

Figure 7. Histograms of the instantaneous stream-wise velocity u: (a) $u_*/u_{*c} = 1.11$; (b) $u_*/u_{*c} = 1.22$ (c) $u_*/u_{*c} = 1.30$. In all the panels: all motions (black), NT state (blue) and T state (red). The black and blue line obviously coincide for $u < 0$.

Table 1
Cumulative Number of Particles That Crossed the Reference Line, $\sum N_Q$

u_*/u_{*c}	T + NT	T	NT	T/(T + NT)	NT/(T + NT)
ΣN_Q					
1.11	6	6	0	100%	0%
1.22	37	37	0	100%	0%
1.3	96	99	−3	103%	−3%
$\sum\lvert\Delta x\rvert$					
1.11	$4.24\cdot 10^3$	$3.56\cdot 10^3$	$6.75\cdot 10^2$	84%	16%
1.22	$2.01\cdot 10^4$	$1.77\cdot 10^4$	$2.42\cdot 10^3$	88%	12%
1.3	$3.86\cdot 10^4$	$3.50\cdot 10^4$	$3.59\cdot 10^3$	91%	9%
$\sum\lvert\Delta y\rvert$					
1.11	$2.31\cdot 10^3$	$1.37\cdot 10^3$	$9.42\cdot 10^2$	59%	41%
1.22	$1.01\cdot 10^4$	$7.01\cdot 10^3$	$3.11\cdot 10^3$	69%	31%
1.3	$1.72\cdot 10^4$	$1.31\cdot 10^4$	$4.01\cdot 10^3$	77%	23%
$\overline{k}$					
1.11	2,901	2,549	352	88%	12%
1.22	14,773	13,462	1,311	91%	9%
1.3	28,127	26,263	1,864	93%	7%
$\overline{N}^m$					
1.11	5.45	2.53	2.92	47%	53%
1.22	21.24	11.92	9.33	56%	44%
1.3	34.73	22.63	12.09	65%	35%
$[u]$					
1.11	12.67	28.24	−0.85		
1.22	16.07	29.77	−1.43		
1.3	19.52	31.04	−2.03		

Note. Sum of the total space traveled in the stream-wise and transverse direction, $\sum\lvert\Delta y\rvert$ *mm* and $\sum\lvert\Delta y\rvert$ *mm*. Mean values of kinetic energy per unit mass, $\overline{k}$ mm^2/s^2. Mean number of moving particles, $\overline{N}^m$, and particle velocity, $[u]$ *mm/s*.

We now focus on two primary variables determining the sediment transport rate: the number of moving particles and the particle stream-wise velocity. The expected value of the solid flowrate per unit width is $[q_s] = \overline{N^m}[u]w_g/A$, where $\overline{N^m}$ is the time average number of moving particles, $[u]$ is the expected value of the particle velocity, w_g is the volume of one particle and A is a reference area of the bed. The number of moving particles increases with transport intensity for both the T and NT states, but the relative contribution of the NT state decreases. The average velocity for the T state is much larger than the average for both states because $[u]^{NT}$ is close to zero (and negative in the present experiments). The mean velocity $[u]^T$ increases for increasing u_*, but much less than linearly; a low increasing rate of particle velocity close to the threshold condition has been previously reported by, for example, Ali and Dey (2019) and Fernandez Luque and van Beek (1976). The increase of $[u]^{T+NT}$ with u_* is stronger, but this is because for progressively higher shear velocity, the proportion of the NT state is smaller, as previously mentioned.

As seen, with a small increase of shear velocity, the sediment transport rate (represented by $\sum N_Q$) increases by one order of magnitude, mostly due to the increase of $\overline{N^m}$ for the T state, consistently with the findings of Radice and Ballio (2008). Ancey et al. (2008) showed that the sediment flowrate is proportional to the sum of the Lagrangian velocities. The PDF of the NT velocity is symmetric with an average $[u]^{NT}$ very close to zero. As a consequence, summing all the possible Lagrangian velocities, we found ≈0: thus, the contribution of the NT state to the sediment transport rate is negligible.

5. Discussion

Considering relevant operational issues, in the present work, we estimated the thresholds δr_t and δd_t based on the statistics of manually identified still particles. This method has the following advantages. First, for the human eye, it is easier to identify stillness rather than motion. Second, it provides some recommendations to anyone wishing to apply our definition in a new experimental campaign, as the thresholds can be determined by preliminary experiments taking videos of some tracers glued on the flume bottom and tracking the (immobile) tracers.

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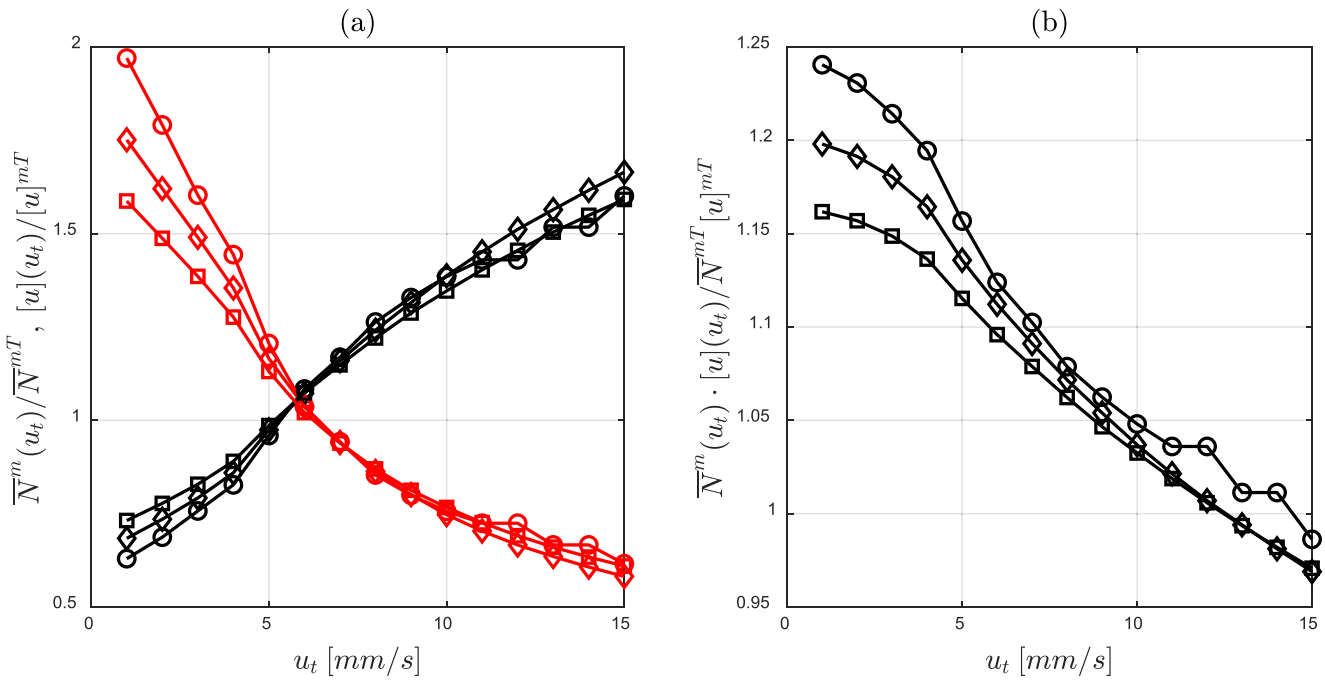

Figure 8. Average number of moving particles as a function of a cut-off velocity $\overline{N^m(u_t)}$ normalized by the expected value for T motion $\overline{N^{mT}}$ (red); Average stream-wise particle velocity as a function of a cut-off velocity $[u](u_t)$ normalized by the expected value for T motion $[u]^T$ (black). (b) Product of $\overline{N^{mT}}$ and $[u](u_t)$, normalized by $\overline{N^{mT}}[u]^T$, as a function of cut-off velocity. $u_*/u_{*c} = 1.11$ (circles), $u_*/u_{*c} = 1.22$ (diamonds) and $u_*/u_{*c} = 1.30$ (squares).

The separation between the T and NT states of motion (either to study both states or to filter the latter to zero, accounting for just the motions that contribute to the sediment flux) is a tricky part of any analysis of sediment kinematics. The criterion used in the present manuscript, once biases induced by experimental uncertainties are properly removed, does not depend on any threshold to be chosen by a user. Furthermore, it is based on instantaneous conditions and not on, for example, integrated displacements or durations of resting times. Our criterion also has some limitations related to being restricted to the stream-wise particle position (a generalization to more complex motion might not be straightforward) and unable to recognize the particle state at the first instant in which a particle is observed because information about the particle at previous time instants is unavailable (thus, if a particle track starts with a forward movement, this is always labeled as in T state). However, the applied separation successfully returned the desired features of the two states: downstream mass transport (state T) and isotropic jiggling (state NT).

Figure 7 demonstrates the existence of a velocity range where the histograms of u for the T and NT states coexist. By contrast, some authors aiming at isolating the motions relevant for sediment flux distinguished the states of stillness and motion imposing a cut-off stream-wise velocity (Ancey & Heyman, 2014; Lajeunesse et al., 2010; Roseberry et al., 2012): here, we reasonably assume that what they considered as motion state corresponds to our T state. Even if a back analysis of prior definitions of motion is not a core objective of the present manuscript, we discuss how a sediment transport variable can be affected by the criterion used for distinguishing the T and NT states. We applied a sharp cut-off velocity criterion and considered a particle in motion state for (a) instantaneous displacement larger than that of our minimum measurable displacement ($\delta r(t) > \delta r_t$) and (b) instantaneous velocity larger than an arbitrary threshold ($u(t) > u_t$). Then, we computed the mean values of the number of moving particles and of the stream-wise velocity component as functions of the threshold u_t: $\overline{N^m}(u_t)$ and $[u](u_t)$. Figure 8a depicts these mean values normalized by the corresponding ones obtained for the T state with our criterion, $\overline{N^{mT}}$ and $[u]^T$. From this figure, it is clear that there is a continuous dependence of $\overline{N^m}$ and $[u]$ on the chosen threshold u_t. Figure 8b shows the result of the same exercise considering the product of the two quantities (that, as stated above, is a proxy for the solid flowrate), and we

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also observe a trend, even if the range of variation is less than in the previous case due to some compensation effect. We observe that, for low u_t, the solid transport rate is overestimated, while the opposite is true for high u_t. We explain these results by considering that the solid flowrate is proportional to the sum of all the velocities. In the previous section, we showed that the sum of all NT velocities is close to zero and that the solid transport rate is due to the sum of T velocity only. If a (positive) threshold velocity is used, the sum of NT velocity is positive and, as a consequence, for low u_t, the solid transport rate is overestimated. On the other hand, for high u_t we are summing only some of the T velocities, and thus will obtain a lower transport rate. In conclusion, the result of the exercise demonstrates that a non-parametric criterion would be preferable to a parametric one.

After doing the exercise of separating motions in the T and NT states, one can judge if the separation is worth being performed. In our opinion, separation is worthwhile from a phenomenological point of view, as it increases the richness of our depiction of the process. Quantitatively, we have demonstrated that the inclusion of the NT state significantly affects the primary variables $\overline{N^m}$ and $[u]$. By contrast, if the primary purpose of a study is the determination of the sediment transport rate, we can disregard the NT state (i.e., considered as stillness); alternatively, it can be simply excluded from the computation of mean values to save some processing time. The contribution of the NT state to the total traveled distance and the particle kinetic energy is non-negligible. These variables furnish, ultimately, an indication of the energy expense needed for the flow to sustain the motion of the solid fraction. Furthermore, the kinetic energy is similar to the granular temperature appearing in approaches based on kinetic theory that is sometimes used to model sediment transport, for example, (Charru et al., 2004).

There are also other problems where our separation among states may furnish new means of analysis. It is well-known that the determination of incipient motion conditions is quite uncertain (Buffington, 1999; Buffington & Montgomery, 1997), due to a continuous transition from stillness to motion rather than an abrupt switch from a stationary bed to bed-load at a certain hydrodynamic condition. In the past, different approaches have been proposed to quantitatively define incipient motion in a repeatable manner: choosing a prescribed low value for the transport rate (Radice & Ballio, 2008), or a low particle activity (Schvidchenko & Pender, 2000). Our present experimental results for weak transport conditions demonstrate that the relative contribution of the state NT to, for example, the number of particles in motion increases for lower discharge. Though we lack experiments closer to a threshold for sediment motion, we can reasonably assume that, for even weaker transport conditions, one would approach relative contributions of the NT state close to 100%. Pähtz et al. (2020) proposed several definitions of a critical Shields parameter. Here, we focus just on the rocking initiation threshold and the initiation threshold. Turbulent fluctuations are strong enough to mobilize the particles in both cases; however, in the second one, the fluctuations are long enough to cause the particles to leave their pockets, whereas in the first case, they do not. The first one can be defined as the probability of observing NT motion greater than zero, while the second one as the probability of observing T motion greater than zero.

In the last part of this discussion, we broaden the scope of the manuscript by considering other processes. We first refer to the challenging problem of finding the elevation of the bed (lower bound of the transport layer) when sediment transport is rather intense (Eslami et al., 2022; Komatsu et al., 2001). The bed elevation can be identified as the elevation at which (a) the vertical profile of the stream-wise sediment velocity attains a nearly zero value (Taberlet et al., 2003), (b) solid concentration reaches a critical value (Berzi et al., 2019) or (c) the production of fluctuating energy is maximum because particle rebounds cause high-velocity correlation (Pähtz & Durán, 2018). However, particles within the stationary bed may also jiggle within cages (Richard et al., 2008) analogous to the surface pockets we mentioned for our application, while particle downstream motion would be prevented by other particles blocked by stress chains. In this context, one could introduce a vertical array of search areas and evaluate the occurrence of NT motions and their relative contribution to the number of moving particles. That is, the transition between a bed-load layer and the stationary bed could also be defined based on the relevance of the NT motions.

Jiggling particles trapped within a stationary bed may also, sometimes, experience, neat downstream motion, as shown, for example, by (Houssais et al., 2016). This alternation between jiggling and neat downstream

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motion is called "creeping" and shares obvious similarities to the bed-load process investigated in this study, apart from creeping flow being substantially slower than bed-load transport. However, the methods introduced in the present study to investigate T and NT motions may be applicable also to the process of bed creeping flow. Considering that, in this case, the instantaneous velocities for the T and NT states would be of the same order of magnitude, it might be harmful to discriminate between motion and rest by using a threshold velocity.

6. Conclusions

In this paper, we conceptualized bed-load particle motion using three states: stillness, transport (T), and non-transport (NT). With reference to recent laboratory experiments, we tracked individual particles, quantified experimental errors introducing objective filters and, finally proposed one-dimensional, instantaneous and, non-parametric criteria for distinguishing the states.

The main conclusions of this work are.

- The proposed criteria clearly allow a distinction between different behaviors of particles whose mass is transported downstream and particles that fluctuate without a significant contribution to mass transport.
- The stream-wise velocity of particles in state T is larger than the transverse one, while the motion of particles in state NT is isotropic.
- The histograms of the stream-wise sediment velocity for the T and NT states partially overlap, thus the two states cannot be separated by criteria based on just a cut-off velocity value.
- Accounting or not accounting for the values for NT state changes the mean values of variables of interest for the bed-load process (number of moving particles, particle velocity, total length traveled by particles, sediment kinetic energy).
- The separation between the T and NT states is worthwhile from a phenomenological point of view. Depending on what one is interested in, the inclusion of the NT state in the computation of mean values can be more or less appropriate. Particularly, we recommend excluding the data for the NT state when determining the sediment transport rate.
- The proposed methodology can be applied to similar processes, such as incipient motion conditions, the definition of solid-fluid interface, and creeping.

Appendix A: Effect of the Temporal Resolution

In this Appendix, we present the effect of temporal resolution on the thresholds involved in the image processing method and on the resulting velocity measurements. For this purpose, we did not perform a new particle tracking, but we analyzed the trajectories once resampled at a lower resolution. Starting from the $f = 32$ Hz, we resampled at the following frequencies: $f = 16$, 10.66, and 8 Hz.

At any temporal resolution, we estimated the thresholds δr_t and δd_t resampling the sub-dataset of still, non-crossed particles already used in the manuscript. Figure A1a shows the thresholds δr_t (in red) and δd_t (in black) as a function of the frame rate: both thresholds are decreasing functions of the frame rate because, if the frame rate is higher, the change of the number of pixels considered as a particle spot is smaller.

Then, we computed the probability distribution functions of the streamwise velocity for all types of motions. The results are reported in Figure A1b. If the frame rate decreases, we can observe two effects: first, some high velocities are filtered; second, the probability of observing small velocities (i.e. the peak around $u = 0$) increases. The explanation of the first finding is straightforward: a reduction of the acquisition frequency is conceptually similar to a moving average, thus extreme values are filtered. Furthermore, the area below a pdf is by definition equal to one. Therefore, if the probability of observing high velocity is reduced, the ones of observing low velocity increase, explaining the second finding.

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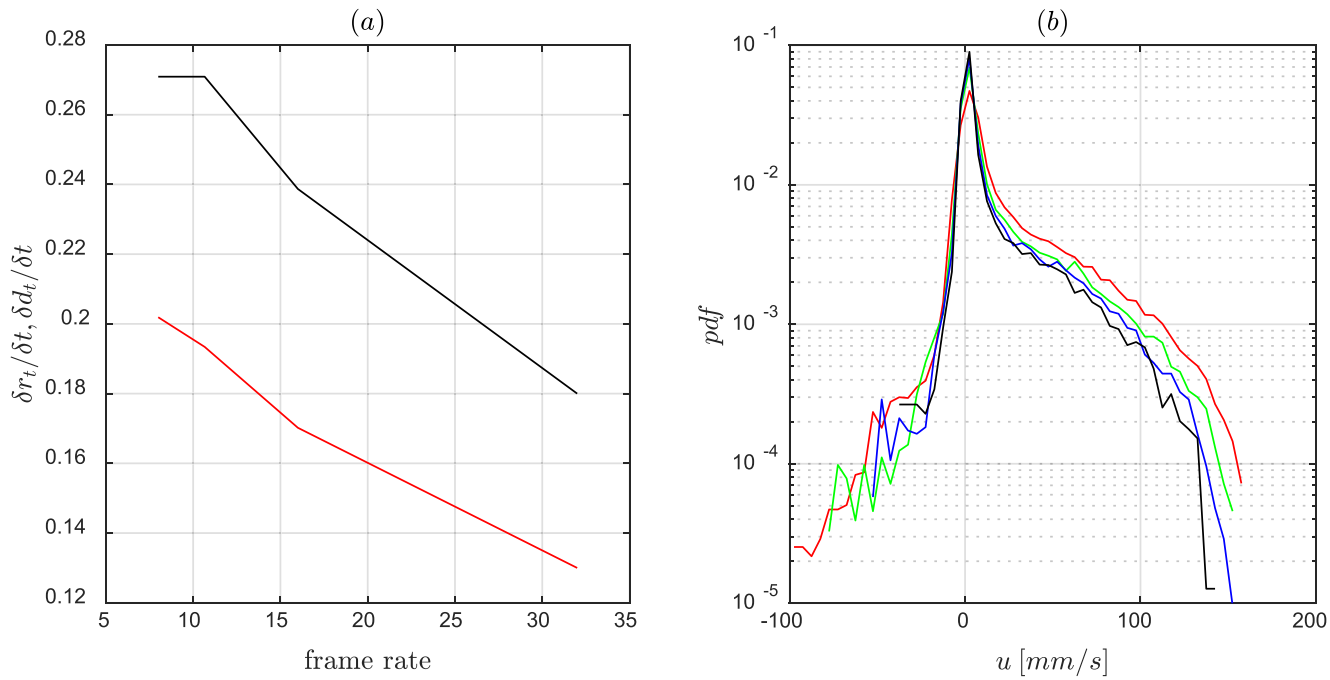

Figure A1. (a) Thresholds δr_t (in red) and δd_t (in black) as a function of the frame rate. Probability distribution function of the streamwise velocity. Red: $f = 32$ Hz. Green: $f = 16$ Hz. Blue: $f = 10.66$ Hz. Black: $f = 8$ Hz.

Appendix B: Comparison With Salevan et al. 2017 Model

In Figure B1 we depict the observed PDFs of a dimensionless particle velocity obtained using the characteristic settling velocity, $V_s = \sqrt{g\Delta d g}$ (where g is the gravitational acceleration, $\Delta = (\rho_s - \rho)/\rho$) and ρ is water density), as a scaling factor, following Lajeunesse et al. (2010). The PDF of the instantaneous particle velocity has been widely studied in the literature, with most of the studies focused on the PDF of u^T. The latter has been proposed to follow an exponential (Fathel et al., 2016; Lajeunesse et al., 2010; Roseberry et al., 2012), normal (Ancey & Heyman, 2014; Heyman, 2014; Martin et al., 2012), Gamma (Liu et al., 2019), lognormal (Shim & Duan, 2019) and alpha stable distribution with power law tails (Zhu et al., 2019). To the best of our knowledge, only Salevan et al. (2017) modeled the PDF for all the motion states, proposing the following bimodal distribution:

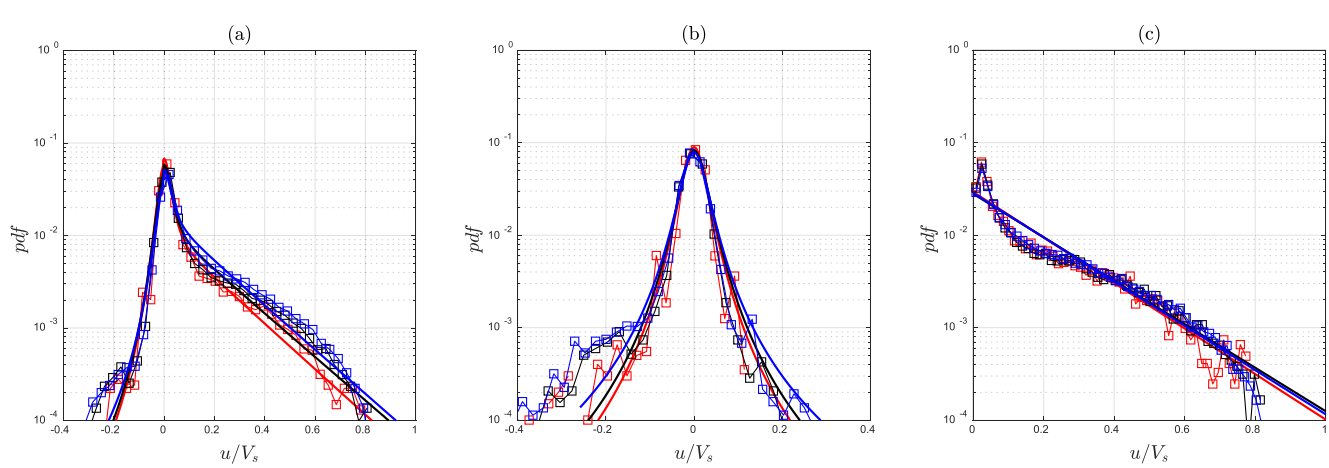

Figure B1. Probability distribution function of the instantaneous velocity u: (a) all motion; (b) NT state; (c) T state. For all the panels: $u_*/u_{*c} = 1.11$ (red), $u_*/u_{*c} = 1.22$ (black) and $u_*/u_{*c} = 1.30$ (blue). Continuous lines represent the theoretical PDFs (Equations B1, B2, and B3), while squares are for the experimental ones. For $u_*/u_{*c} = 1.11$: $A = 0.63$; $\sigma = 4.04$; $\xi = 2.64$; $\mu_u = 34.12$. For $u_*/u_{*c} = 1.22$: $A = 0.56$; $\sigma = 4.36$; $\xi = 2.51$; $\mu_u = 35.72$. For $u_*/u_{*c} = 1.30$: $A = 0.45$; $\sigma = 4.51$; $\xi = 2.12$; $\mu_u = 35.10$.

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$$f(u \mid A,\ \sigma, \xi, \mu_u) = Af_1(u \mid \sigma, \xi) + (1-A)f_2(u \mid \mu_u) \tag{B1}$$

where:

$$f_1(u^{NT} \mid \sigma, \xi) = \frac{\Gamma\left(\frac{\xi+1}{2}\right)}{\sigma\sqrt{\xi\pi}\Gamma\left(\frac{\xi}{2}\right)}\left(\frac{\xi + \left(\frac{u^{NT}}{\sigma}\right)^2}{\xi}\right)^{-\frac{\xi+1}{2}} \tag{B2}$$

$$f_2(u^T \mid \mu_u) = \frac{1}{\mu_u}\exp - \frac{u^T}{\mu_u} \tag{B3}$$

This model is composed by two terms describing the behavior of the NT (f_1) and T states (f_2). We calibrated the parameters of (A1) considering all the moving particles. Figure A1a shows a satisfactory comparison between the experimental PDF and the theoretical distribution. In the caption we report the calibrated parameters: the values of μ_u are consistent with those of Salevan et al. (2017); the other parameters are different but trends are preserved (for increasing u_*/u_{*c}, σ and A increase, while ξ decreases). The detected discrepancy may be due to the many small displacements considered by Salevan et al. (2017); in fact, the peak of their distribution was one order of magnitude bigger than the one showed in Figure A1a. We remark that in this work many small displacements were filtered out by the thresholds δr_t and δd_t; in addition, our frame rate was smaller than that of Salevan et al. (2017) (32 vs. 250 fps), thus causing additional filtering, as shown in Appendix A.

In their work, Salevan et al. (2017) did not propose any criteria for the distinction between the T and NT states. In contrast, in this manuscript we did so, and thus we were able to obtain the experimental PDF of the particle velocity for both states. Without any further calibration, in Figures A1b and A1c we compare the experimental and the theoretical distributions of the two states, separately. The agreement is satisfactory, even though the experimental PDF for the NT state is slightly skewed left.

Notation

A	model parameter
a_x	instantaneous particle x-acceleration
a_y	instantaneous particle x-acceleration
d	sediment diameter
g	acceleration due to gravity
k	particle kinetic energy per unit mass
M^m	clipping function for the motion state
M^s	clipping function for the still state
M_Q	clipping function for the particles crossing a specific line
N^m	number of particles in motion state at time t
N_Q	number of particles that crossed a specific line at time t
u	instantaneous particle x-velocity
u_*	shear velocity
u_{*c}	critical shear velocity
v	instantaneous particle y-velocity
V	velocity magnitude

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V_s	characteristic settling velocity $\left(V_s = \sqrt{g\Delta d}\right)$
x	streamwise coordinate
x_*	transversal line where mass transport is measured
X^T	any variable related to the transported particles
X^{NT}	any variable related to the non-transported particles
$\overline{X}$	any variable averaged in time
y	transversal coordinate
δd	change of the measured diameter between two consecutive frames
δd_t	threshold for spot diameter change
δr	instantaneous displacement
δr_t	minimum measurable displacement
δt	small time interval
δx	instantaneous x-displacement
δy	instantaneous y-displacement
Δ	sediment density ratio ($\Delta = \rho_s/\rho - 1$)
μ_u	model parameter
ξ	model parameter
ρ	water density
ρ_s	sediment density
σ	model parameter

Data Availability Statement

The particle's trajectories used in the study are available in Rebai et al. (2023).

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